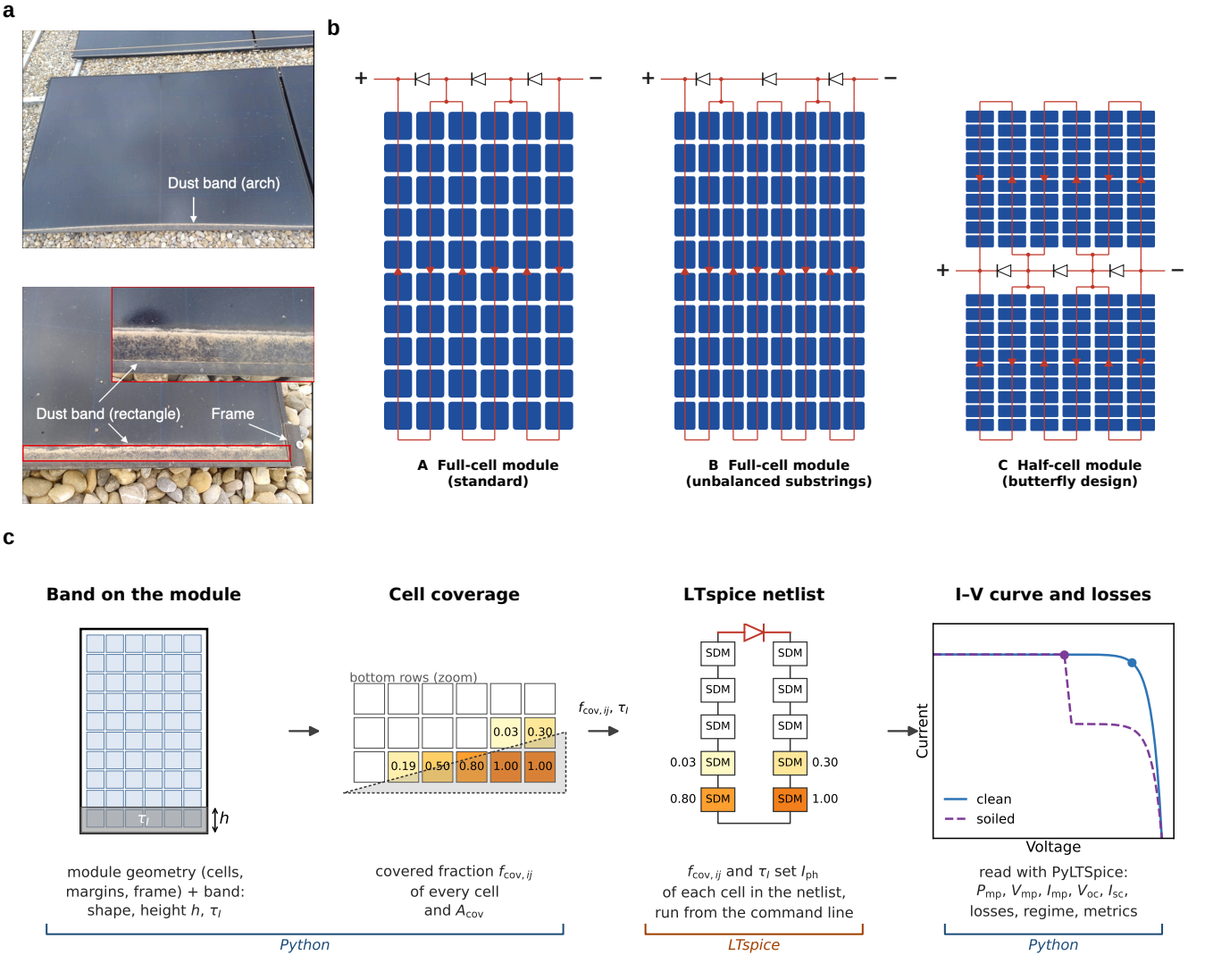

a
Dust band (arch)
Dust band (rectangle)
Frame
b
A Full-cell module (standard)
B Full-cell module (unbalanced substrings)
C Half-cell module (butterfly design)
c
Band on the module
Cell coverage
LTspice netlist
I-V curve and losses
h
bottom rows (zoom)
0.03
0.30
0.19
0.50
0.80
1.00
1.00
SDM
Current
Voltage
clean
soiled
module geometry (cells, margins, frame) + band: shape, height h, τl
covered fraction fcov, ij of every cell and Acov
fcov, ij and τl set Iph of each cell in the netlist, run from the command line
read with PyLTSpice: Pmp, Vmp, Imp, Voc, Isc, losses, regime, metrics
Python
LTspice
Python